Real-Time Mental Health Monitoring AI

Emotion Detection for Astronauts Using Deep Convolutional Neural Networks (CNNs)

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BACKGROUND RESEARCH

Psychological & Physiological

Warning signs of mental health issues:

- Changes in sleep or appetite
- Mood fluctuations
- Withdrawal or apathy
- Decline in cognitive function, memory, or concentration
- Increased sensitivity to stimuli
- Anxiety or unusual behaviors

NASA's Project Mercury

- Body temperature slightly increased
- Blood pressure increased
- Respiration was higher at lift-off

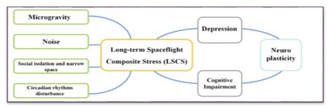


Figure 2 - Long-term Spaceflight Composite Stress [1]

AI Models

A **FER Human-Robot** study used a two-stage CNN pipeline, a neural network that first detects & aligns faces which the second then uses to analyze emotions. The names of the two different neutral networks that were used were:

ResNet Inception Based Faster R-CNN Model
and Inception Based SSD Model.

- Both had about the same total loss and accuracy rate
- Accuracy rates were ~83%

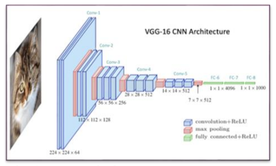


Figure 3 – Convolution Neural Networks from OpenCV [2]

Emotion Recognition

- Emotion detection using voice data and analyzing speech data
- Dataset included 40 commands in five emotions (angry, happy, sad, bored, and neutral) and had 14 speakers (7 male, 7 female)
- Achieved an 80% accuracy rate

Measuring Depression using Body Movements AI

- Study used 40-60 minute recordings of clinically diagnosed patients of Major Depressive Disorder (MDD)
- Model prediction and accuracy was compared to patients' self-reported scales
- Model was trained on 2013 FER dataset and resulted In a 70.71% accuracy rate

RESEARCH OBJECTIVES

1. **CHALLENGE STATEMENT –**
   - On long duration space missions, the psychological health of the astronauts is a critical issue. Assessing the mood and emotions of the crew member can improve astronaut performance and mission success.
2. **RESEARCH QUESTION –**
   - How can the use of AI track the psychological impacts of isolation, stressful missions, and physical implications of a microgravity environment through assessing astronauts' emotions using onboard softwares?
3. **ETHICAL CONSIDERATIONS –**
   - Racial and gender diversity
   - Psychological monitoring, will not be shared with third parties
   - No personally identifiable information linked to dataset
4. **EXPECTED OUTCOMES –**
   - AI model that detects emotions with 60% - 80% accuracy
   - Real-time monitoring tool (i.e. a Streamlit app) to provide feedback to the model
5. **POTENTIAL CHALLENGES –**
   - Skew in the dataset creates bias
   - Large amount of computational resources, may be limited on spacecrafts
   - Fluid shifts in microgravity alters facial appearance,
   - Overfitting and underfitting, model performs well on dataset but poorly in real-world scenarios
   - Image lighting and quality may impact the model's performance

ENGINEERING METHODOLOGY

Select a Dataset → Data Preprocessing → Build the Model → Train & Test Model → Build App

Research Datasets on:
- Kaggle
- Hugging Face
- Papers with Code
- Github

Datasets:
- Google API Open Datasets
- Kaggle – FER 2013 Dataset
- Chicago Face Database (CFD)
- Celebrity Voice Datasets
- RAVDESS

Visualize images from the dataset and plot into a graph using PyPlot

Visualize audio from the dataset and plot into a graph using PyPlot and Librosa. And visualize audio spectrogram

Design the model architecture through creating layers which process input data.
Find trainable parameters and evaluate model by measuring its accuracy score (compares predicted vs. actual labels to assess effectiveness).

Split images into train and test variables.

Train the model multiple times and over several epochs to fine-tune performance, optimize hyperparameters, reduce overfitting, adapt to new data, and improve accuracy.

Create a Streamlit app powered by ngrok to demonstrate real-time emotion prediction using user-captured images. Correct predictions validate the model's accuracy; incorrect ones are added to a dataset predictions are used for retraining, improving performance over time through continuous learning.

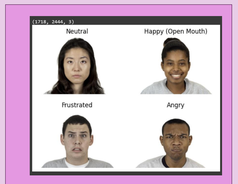
Face Emotion Recognition (FER)

Chosen dataset:

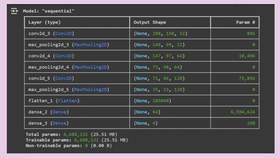
Chicago Face Database (CFD)
4 different emotions: Happy, Neutral, Frustrated, Angry

Programming Language: Python

Programming Environment: Google Colab



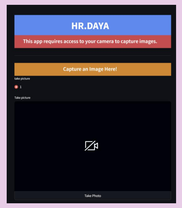
Through data preprocessing, the images were resized to (300, 200)



Layer (type)	Output Shape	Param #
conv2d_0 (Conv2D)	(None, 298, 198, 32)	896
max_pooling2d_0 (MaxPooling2D)	(None, 149, 99, 32)	0
conv2d_1 (Conv2D)	(None, 147, 97, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 73, 48, 64)	0
conv2d_2 (Conv2D)	(None, 71, 46, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 35, 23, 128)	0
flatten_1 (Flatten)	(None, 103040)	0
dense_2 (Dense)	(None, 64)	6,594,624
dense_3 (Dense)	(None, 4)	260

Total params: 6,688,132 (25.51 MB)
Trainable params: 6,688,132 (25.51 MB)
Non-trainable params: 0 (0.00 B)

All parameters (features) from this dataset are trainable (6,688,132 out of 6,688,132 are trainable)



Speech Emotion Recognition (SER)

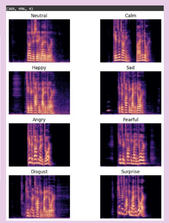
Chosen dataset:

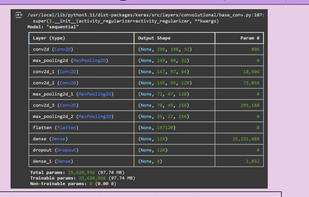
RAVDESS Dataset
8 different emotions: Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, and Surprise

Programming Language: Python

Programming Environment: Google Colab

Visualized waveform and spectrogram graphs of audio. Convert all audio files into image form to train the model which is CNN based

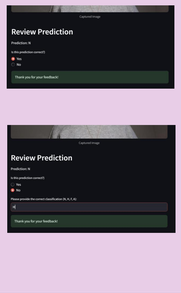




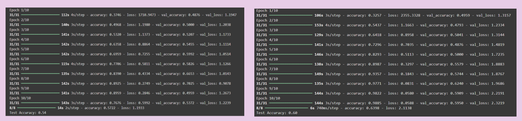
Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 128, 256)	896
max_pooling1d (MaxPooling1D)	(None, 102, 32)	0
conv1d_1 (Conv1D)	(None, 102, 64)	18,496
conv1d_2 (Conv1D)	(None, 102, 128)	73,856
max_pooling1d_1 (MaxPooling1D)	(None, 51, 128)	0
conv1d_3 (Conv1D)	(None, 51, 256)	295,168
max_pooling1d_2 (MaxPooling1D)	(None, 25, 256)	0
flatten (Flatten)	(None, 10240)	0
dense (Dense)	(None, 128)	25,251,488
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 5)	1,810

Total params: 25,640,936 (97.79 MB)
Trainable params: 25,640,936 (97.79 MB)
Non-trainable params: 0 (0.00 B)

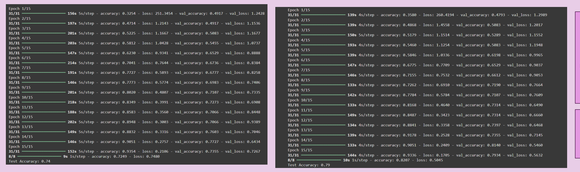
All parameters (features) from this dataset are trainable (25,620,936 out of 25,620,936 are trainable)



RESULTS



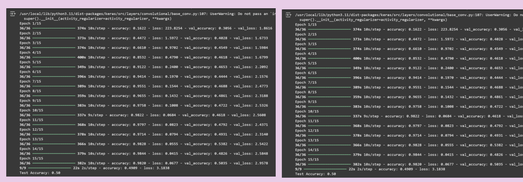
FER model was training over 10 epochs with a 54% and 60% accuracy score



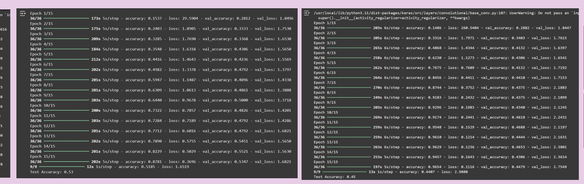
FER model was training over 10 epochs with a 74% and 79% accuracy score

FER model was trained 4 times. Two trainings for the model with 10 epochs and two trainings with 15 epochs

Highest accuracy percentage was 83% with 15 epochs



SER model was training over 15 epochs with a 50% accuracy score for both instances

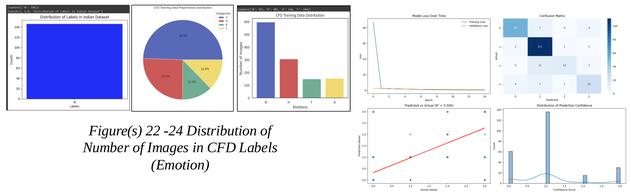


SER model was training over 15 epochs with a 53% and 45% accuracy score

SER model was trained 4 times. All trainings were done over 15 epochs

Highest accuracy percentage was 53% with 15 epochs

DATA ANALYSIS



Figure(s) 22 -24 Distribution of Number of Images in CFD Labels (Emotion)

Figure 25 - FER Training and Validation Loss Graphs

Figure 25 displays training and validation loss over 15 epochs for a model with an 83% accuracy rate, a confusion matrix, prediction accuracy, and confidence distribution. In *Model Loss Over Time* (top-left), the losses converge, showing no severe overfitting. In *Confusion Matrix* (top-right), the diagonal values (54, 111, 13, 23) indicate correct classifications, while off-diagonal values highlight misclassifications. In the *Predicted vs. Actual* graph, the red line represents an ideal correlation, and the spread of points suggests moderate accuracy ($R^2 = 0.5$, $R^2 = 0.5$, $R^2 = 0.5$), meaning the model explains 50% of the variance. Peaks around 0, 1, and higher values indicate varying confidence levels, while the density line provides a smooth estimate of prediction confidence distribution.

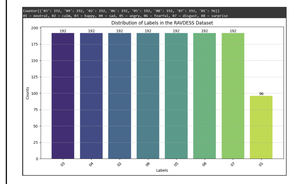


Figure 26 - Distribution of Number of Images in RAVDESS AUDIO Labels (Emotion)

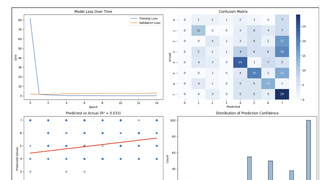


Figure 27 - SER Training and Validation Loss Graphs

Figure 27 displays training and validation loss over 15 epochs, a confusion matrix, prediction accuracy, and confidence distribution. In Model Loss Over Time (top-left), the loss decreases rapidly in early epochs but plateaus, showing possible underfitting. In Confusion Matrix (top-right), the diagonal values represent correct classifications, while off-diagonal values indicate misclassifications, showing inconsistencies in predictions. In Predicted vs. Actual (bottom-left), the red regression line suggests a weak correlation ($R^2 = 0.033$), indicating the model explains very little variance in actual values. In Distribution of Prediction Confidence (bottom-right), peaks around specific values suggest confidence clustering, with a density line providing a smooth estimate of prediction confidence.

CONCLUSION

- Dataset had a high frequency of neutral images
- Highest accuracy for FER model was 83% with 15 epochs of training
- Training and Validation Loss was high meaning the model may perform well on its own training data instead of on applicable real world scenarios
- Improve model accuracy for both models and decrease model training and validation loss

APPLICATIONS

- Helps clinicians monitor emotional changes in patients with mood disorders
- Helps with remote therapy/telehealth (facial expressions are used to tell a patient's emotional state)
- Serves as a screening tool to detect early signs of mental health conditions such as depression PTSD, and anxiety
- Supports patients who have a hard time communicating (speech-impediment, autism, selective-mutism, and special education students)
- Can supports Alzheimer's and dementia patients to improve care and treatment

FUTURE WORK

Add SER model to Streamlit web application

Improve Overall Both Model Accuracy:
- Using different neural networks to improve model accuracy (such as VGG16)

Pupil Dilation:
- Evaluating tell signs of eyes/pupil dilation

Different Ways to Evaluate Mental Health:
- Testing blood pressure to depict stress levels